

RESEARCH

Open Access



Litterfall dynamics and decomposition-driven nutrient cycling in sacred forests of Eastern India

Antaryami Pradhan¹ , Abinash Mansingh^{2*} , Jadhav Sudarshan Gopinath³ and Nirius Jenan Ekka²

*Correspondence:

Abinash Mansingh

abinashmansingh@suniv.ac.in

¹Terracon Ecotech Private. Limited,
Mumbai 400054, India

²School of Life Sciences, Sambalpur
University, Burla, Odisha
768019, India

³Department of Forest,
Environment & Climate Change,
Government of Odisha,
Sambalpur 768001, India

Abstract

Litterfall and decomposition are pivotal processes governing nutrient cycling and ecosystem functioning, yet they remain poorly studied in India's sacred forests. This study quantified litterfall production, standing litter biomass, decomposition, and nutrient return in four tropical dry deciduous sacred forests of Western Odisha, India, over one year (January–December 2023). The study sites comprised *Shorea robusta*-dominated forests at Andhari, Dedungri, and Medha, and a *Cleistanthus collinus*-dominated forest at Papanga. Litterfall was collected using 1 m² litter traps, and decomposition was assessed using 20 g litterbags with 1 mm mesh. Litter fractions were analyzed for carbon (C), nitrogen (N), phosphorus (P), and potassium (K). Annual litterfall ranged from 8.59 ± 0.79 Mg ha⁻¹ yr⁻¹ at Andhari to 11.32 ± 1.12 Mg ha⁻¹ yr⁻¹ at Medha, with leaves contributing 80.65–83.35% of total litterfall. Standing litter biomass ranged from 2.39 ± 1.27 Mg ha⁻¹ at Andhari to 2.68 ± 1.42 Mg ha⁻¹ at Medha. Litter turnover rates (K_i) ranged from 3.59 to 4.22 yr⁻¹, corresponding to residence times of 87–102 days. Decomposition was rapid, with approximately 95% mass loss within six months; decay rate constants (k) ranged from 5.75 to 6.65 yr⁻¹, with half-lives (t₅₀) of 38–44 days and 99% decomposition (t₉₉) occurring within 274–318 days. Nutrient release followed the order K > N > P, with Papanga exhibiting the highest nutrient loss (96.57%) after six months. Initial litter N and P concentrations were significantly correlated with decomposition rates, while K declined rapidly due to leaching. These findings highlight the critical role of sacred forests in nutrient cycling and ecosystem functioning and provide valuable insights for their conservation under increasing anthropogenic pressures.

Keywords Litterfall dynamics, Decomposition processes, Sacred forests, Nutrient cycling, Tropical dry deciduous forests, Ecosystem functioning

1 Introduction

Sacred groves are traditionally protected forest patches conserved by local communities through religious beliefs and cultural practices, representing one of the oldest forms of in situ biodiversity conservation in India. These ecologically sensitive landscapes support rich biodiversity and contribute to microclimate regulation, soil moisture conservation,



© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

and nutrient cycling. In forest ecosystems, leaf litter forms a crucial link between vegetation and soil processes, serving as a major source of organic matter and nutrients [1]. In sacred groves, where anthropogenic disturbances are generally limited due to restrictions on timber and firewood extraction, NTFP removal, grazing, and other resource-use activities, litterfall and its subsequent decomposition are important components of ecosystem functioning and may contribute significantly to soil fertility and ecological resilience [2, 3]. Understanding litter dynamics in these culturally protected ecosystems can therefore provide valuable insights into their role in sustaining essential ecosystem processes and maintaining environmental health [4].

Litterfall is a fundamental ecological process that links primary production with nutrient cycling in terrestrial ecosystems. It represents a major pathway through which organic matter and nutrients are transferred from vegetation to the soil, where litter decomposition regulates nutrient availability for plant growth [5, 6]. The dynamics of litterfall and its subsequent decomposition are key determinants of soil fertility, carbon storage, and forest productivity, with important implications for climate change mitigation. In tropical dry forests, nutrient cycling is strongly influenced by seasonal water availability and nutrient status, both of which regulate litter production and decomposition rates [7–10]. Seasonal variation in rainfall and plant phenology can alter litterfall patterns, while plant functional traits, including leaf quality and structural characteristics, influence litter decomposability and nutrient release [11–14]. The decomposition process is further regulated by soil microbial communities, whose activity and composition respond to moisture conditions and litter quality, thereby influencing nutrient availability, carbon dynamics, and overall ecosystem functioning [15–17].

Despite their ecological significance, research on litterfall and decomposition in India's tropical dry deciduous forests remains limited, particularly in sacred forests. These forests are traditionally conserved patches of climax vegetation, protected through religious beliefs and cultural practices. They function as biodiversity hotspots and provide essential ecosystem services, including carbon sequestration, water regulation, and soil conservation, which are especially important in human-dominated landscapes [18–20]. Although their biodiversity and conservation values have been widely recognized, their role in ecosystem processes such as litter decomposition and nutrient cycling has received comparatively little attention [21–23]. A better understanding of litter dynamics and associated nutrient release in sacred forests is essential for elucidating their contribution to ecosystem functioning, nutrient cycling, and ecological resilience under increasing anthropogenic pressures.

Tropical dry deciduous forests of Eastern India, including those of Western Odisha, exhibit pronounced seasonality in litterfall, primarily driven by dry-season water stress that induces leaf abscission [24–26]. Variations in soil water availability and plant water-use strategies further influence vegetation structure, productivity, and nutrient cycling, underscoring the ecological importance of seasonal water stress in these ecosystems [27]. Litter decomposition is regulated by litter biochemical composition, microbial activity, and climatic conditions [28–32]. Although litter dynamics have been extensively documented in tropical evergreen and semi-evergreen forests [33–36], comparable information for sacred forests, particularly tropical dry deciduous sacred forests, remains limited. The role of small, culturally protected, and relatively undisturbed sacred forests in regulating litter decomposition and nutrient fluxes is still poorly understood,

despite their potential importance as localized reservoirs of organic matter and nutrients in human-dominated landscapes [37–41]. This represents a significant gap in the current understanding of ecosystem functioning in tropical dry deciduous sacred forests.

The novelty of this study lies in providing region-specific information on seasonal litter production, decomposition, and nutrient dynamics in the sacred forests of Western Odisha, a region for which such ecological information remains limited. Although litterfall and nutrient cycling have been extensively investigated in tropical forests and, to a lesser extent, in sacred groves and sacred natural sites in India and elsewhere, comparable information on litter decomposition and nutrient fluxes in the sacred forests of Western Odisha remains scarce [42–46]. By quantifying litterfall production, standing litter biomass, decomposition rates, and associated C, N, P, and K fluxes over a one-year period (January–December 2023), this study provides baseline information on nutrient cycling in these culturally protected ecosystems. The specific objectives were to: (i) quantify litterfall production and standing litter biomass, (ii) assess litter decomposition rates and associated nutrient release, and (iii) evaluate nutrient fluxes across sacred forests. The findings provide new insights into the functional role of sacred forests in nutrient cycling and ecosystem functioning and may support evidence-based conservation and management strategies for these ecologically and culturally significant ecosystems in rapidly changing landscapes [47–51].

2 Methodology

2.1 Study area

The present study was conducted in four sacred forests located in the western region of Odisha, India, in a tropical dry deciduous forest ecosystem. These forests are part of the unique socio-ecological landscapes in Odisha, where sacred forests are preserved by local communities due to cultural and religious significance. The selected sites for this study include Andhari, Dedungri, Medha, and Papanga; present in Jharsuguda, Sambalpur, Sundargarh and Bargarh district respectably. Each representing distinct sacred forest ecosystems that varies in terms of species composition, forest structure, and proximity to human activities. The geographical region lies approximately between 20° 40' N to 22° 32' N Latitude and 82° 39' E to 85° 22' E Longitude. A Digital Elevation Model (DEM) showing the spatial distribution of the sacred forests is presented in Fig. 1, where elevation is expressed in meters above mean sea level (AMSL). The elevation of the study sites ranges approximately from 180 to 310 m above sea level.

The study area is characterized by a tropical savanna climate, with a distinct seasonal pattern of temperature and rainfall (Fig. 2), which plays a significant role in shaping the vegetation structure and litter dynamics of the forests. The mean annual temperature ranges from 22 °C to 30 °C, with the hottest months being April to June, during which maximum temperatures may exceed 45 °C. The monsoon season (June to September) contributes to the majority of the annual rainfall, with total annual precipitation ranging from 900 mm to 1,400 mm, depending on the specific location (Source: IMD, Annual climate summary report 2023).

The presence of these sacred forests within human-dominated landscapes provides a unique opportunity to investigate their role in sustaining ecological processes such as litterfall production, decomposition, and nutrient cycling despite increasing anthropogenic pressures. The vegetation is characteristic of tropical dry deciduous forests and is

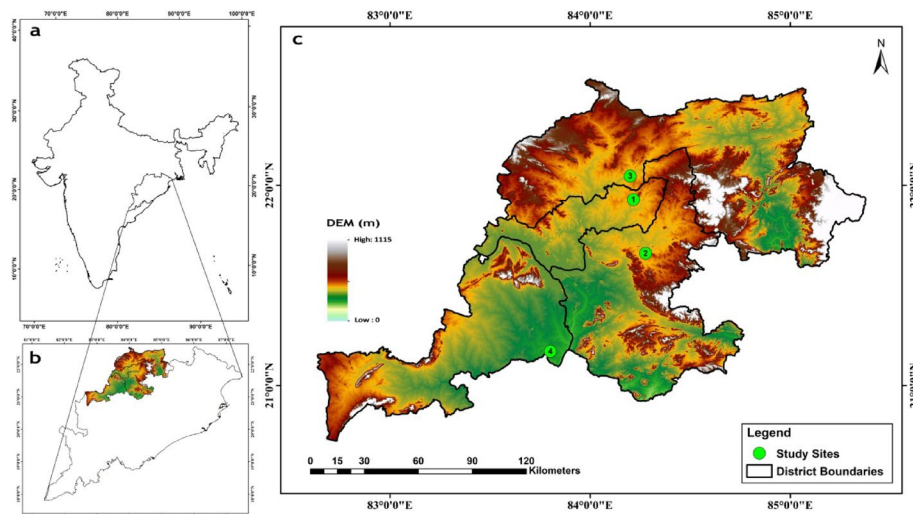


Fig. 1 Geographic location of the study area showing **a** the position of Odisha within India, **b** the study region within Odisha, and **c** the spatial distribution of the selected sacred forest sites: Andhari (1), Dedungri (2), Medha (3), and Papanga (4)

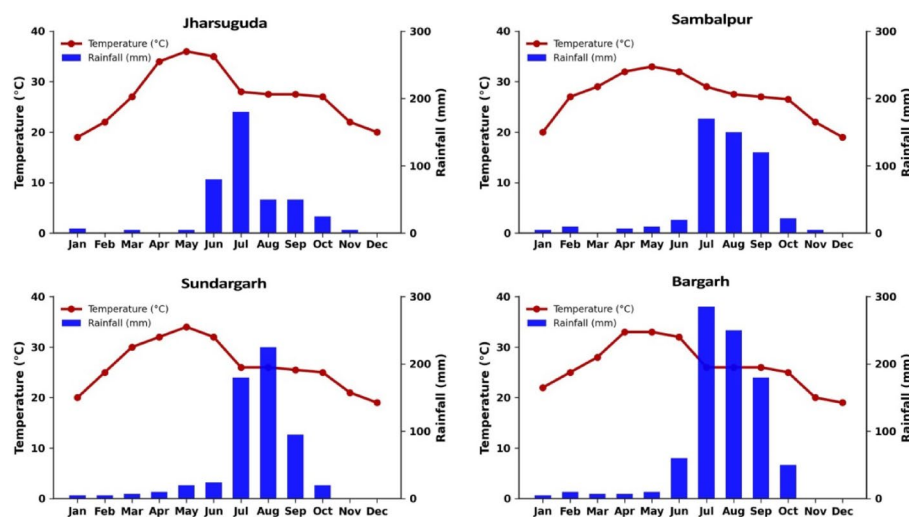


Fig. 2 Mean monthly air temperature and rainfall in the study district of western Odisha during 2023

dominated by species such as *Shorea robusta*, *Cleistanthus collinus*, *Terminalia tomentosa*, *Madhuca longifolia*, and *Buchanania cochinchinensis*, all of which are well adapted to the region's seasonal climate [52]. The ecological characteristics and conservation status of these sacred forests are closely linked to their cultural protection and traditional community-based management practices. Historically, they have been maintained as culturally protected forest patches within increasingly human-dominated landscapes, although the intensity of anthropogenic influence varies among sites. Potential pressures include fuelwood collection, livestock grazing, selective extraction of non-timber forest products, and other localized human activities [3]. The selected forests represent different stages of ecological succession and forest degradation, with Medha and Papanga being relatively well preserved, whereas Andhari and Dedungri exhibit signs of moderate degradation [3]. These differences in forest condition provide a useful framework for

examining spatial variation in litterfall dynamics, decomposition, and nutrient cycling across the sacred forests of the region.

2.2 Methodology

2.2.1 Litter sampling and analysis

Litterfall was quantified at monthly intervals over a one-year period. Within each sacred forest, fifteen litter collection pits (1 m × 1 m × 0.15 m; length × breadth × depth) were randomly established, with a minimum spacing of approximately 100 m between pits to capture spatial variation in species composition and litter inputs across the forest. All pits were located at least 10 m inside the forest boundary to minimize potential edge effects. Each pit was excavated directly into the forest floor, and naturally deposited litter within the designated area was collected during each monthly sampling. No suspended litter traps, plastic sheets, or other artificial collection materials were used. The collected litter was separated into four fractions: leaves, twigs, branches, and other plant parts. Each fraction was oven-dried at 80 °C for 48 h, or until a constant weight was attained, and the dry biomass was recorded.

2.2.2 Litter decomposition

Leaf litter decomposition was studied using the nylon bag method [53]. Fresh leaf litter was collected randomly from all four sites during the peak litterfall period (February–March) and air-dried. Subsamples (20 g) were placed into litterbags (20 cm × 20 cm; mesh size 1 mm × 1 mm). Ninety litterbags were prepared for each site and placed on the soil surface beneath the litter layer in a completely randomized design. Six litterbags were retrieved monthly from each site. Samples were carefully washed to remove soil particles, oven-dried at 80 °C for 48 h, weighed, and powdered for chemical analysis.

2.2.3 Data analysis of litterfall and decomposition

The annual mass of each litterfall component was obtained by summing monthly values, while monthly litterfall was expressed as the mean of all litter traps per site. Litter turnover rate (K_L) was calculated following [54]:

$$K_L = \frac{L}{X_L}$$

where L is annual litterfall ($\text{kg ha}^{-1} \text{ yr}^{-1}$) and X_L is the mean annual standing litter (kg ha^{-1}). Turnover time (T) was calculated as the reciprocal of turnover rate:

$$T = \frac{1}{K_L}$$

The decay constant (k) for leaf litter was computed using the negative exponential decay model of [55]:

$$\frac{X}{X_0} = \exp(-kt)$$

where X is the remaining weight at time t , X_0 is the initial weight, k is the decay rate coefficient, and t is time. The times required for 50% (t_{50}) and 99% (t_{99}) mass loss were calculated as per [56]:

$$t_{50} = \frac{0.693}{k}$$

$$t_{99} = \frac{5}{k}$$

Nutrient return was calculated as the sum of monthly returns. Nutrient concentration of decomposing litter was estimated as:

$$\% \text{ Nutrient remaining} = \left(\frac{C}{C_0} \right) \times \left(\frac{DM}{DM_0} \right) \times 100$$

where C is nutrient concentration in litter at sampling time, C_0 is initial nutrient concentration, DM is litter dry mass at sampling, and DM_0 is initial dry mass [56].

2.2.4 Chemical analysis

All litter species were analyzed together as mixed-species litter to represent the natural litter layer beneath the mixed-species canopy and to reflect site-level litter decomposition and nutrient cycling conditions. Leaf litter collected from litter traps and decomposition bags was carefully cleaned to remove adhering soil particles, washed with distilled water, oven-dried at 80 °C for 48 h, finely ground, and stored for chemical analysis. Ash content was determined by igniting 1 g of powdered litter at 550 °C for 6 h in a muffle furnace, and total carbon was estimated as 50% of the ash-free dry mass [57]. This approach provides an approximate estimate of carbon content and may not fully capture variation in the carbon concentration among different litter types compared with direct elemental analysis. For nitrogen estimation, finely ground samples were digested with concentrated H_2SO_4 using Kjeltabs as a catalyst, and total nitrogen was determined by the Kjeldahl distillation and titration method, with blanks and standards included following the protocol of Allen et al. (1974) [58]. Phosphorus concentration was determined using the molybdenum-blue method [58] after digestion with a tri-acid mixture (H_2SO_4 - HClO_4 - HNO_3). Potassium concentration was measured by flame photometry [59] following digestion with the same tri-acid mixture. Structural components of litter, including lignin and cellulose, were not analyzed in the present study.

2.2.5 Statistical analysis

For litter decomposition, six litterbags were retrieved from each of the four study sites at each monthly sampling interval, resulting in 24 litterbag samples per month ($n=24$). Each litterbag was treated as an independent experimental unit within each sampling period, while sampling month and forest site were considered temporal and spatial factors, respectively. Data were not transformed prior to analysis. Normality and homogeneity of variance were evaluated using the Shapiro-Wilk and Levene's tests, respectively, implemented in the *stats* package in R. When the assumptions of parametric tests were not satisfied, the Kruskal-Wallis rank-sum test was used, without post-hoc pairwise comparisons. For datasets meeting the assumptions of parametric analysis, two-way analysis of variance (ANOVA) was performed to evaluate the effects of forest site, sampling month, and their interaction. Pearson's correlation analysis was used to examine relationships between monthly litterfall and climatic variables, as well as between litter mass loss and selected litter biochemical traits. Correlation analyses were conducted using the *corrplot* package in R. Statistical significance was assessed at $\alpha = 0.05$.

for all analyses. Because repeated observations were collected from the same forest sites over time, the analyses should be interpreted as evaluating site and temporal variation within the study period rather than fully accounting for temporal dependence among observations.

3 Results

3.1 Dynamics of litterfall production

The monthly litterfall production over the 12-month study period exhibited a distinct seasonal pattern, with peaks occurring in March across all four sacred forests (Fig. 3). Although monthly litter production did not differ significantly among the sites ($p=0.3191$), significant variation was observed in the monthly litterfall pattern within sites ($p<2.2\text{e-}16$). Annual litterfall was highest in Medha ($11.32\text{ Mg ha}^{-1}\text{ yr}^{-1}$), followed by Dedungri ($10.28\text{ Mg ha}^{-1}\text{ yr}^{-1}$), Papanga ($8.94\text{ Mg ha}^{-1}\text{ yr}^{-1}$), and lowest in Andhari ($8.59\text{ Mg ha}^{-1}\text{ yr}^{-1}$). Leaves constituted the largest fraction of litterfall, ranging from 80.65% in Medha to 83.35% in Andhari, followed by twigs, branches, and other miscellaneous components. Thus, the order of litterfall fractions across all sites was leaves > twigs > branches > others. However, no formal statistical correlation between monthly litterfall and climatic variables was conducted in the present study; therefore, the observed seasonal association is interpreted descriptively.

3.2 Standing litter and turnover rate

Standing litter exhibited pronounced seasonal fluctuations that closely followed litterfall dynamics, with maximum accumulation occurring in March across all study sites and minimum values during the rainy and post-rainy months (August–November). Standing litter did not differ significantly among forest sites (ANOVA: $F=0.157$, $p=0.925$; Kruskal-Wallis: $\chi^2 = 0.836$, $df=3$, $p=0.841$), whereas significant seasonal variation among months was observed (ANOVA: $F=101.757$, $p<2.2\times 10^{-16}$; Kruskal-Wallis: $\chi^2 = 44.625$, $df=11$, $p<0.001$) (Table 1). Mean annual standing litter biomass was highest in Medha (2.68 Mg ha^{-1}), followed by Dedungri (2.51 Mg ha^{-1}), Papanga (2.47 Mg ha^{-1}), and Andhari (2.39 Mg ha^{-1}). The litter turnover rate (K) ranged from 3.59 yr^{-1} in Andhari to 4.22 yr^{-1} in Medha, corresponding to turnover times of 0.24–0.28 years. Litter residence time

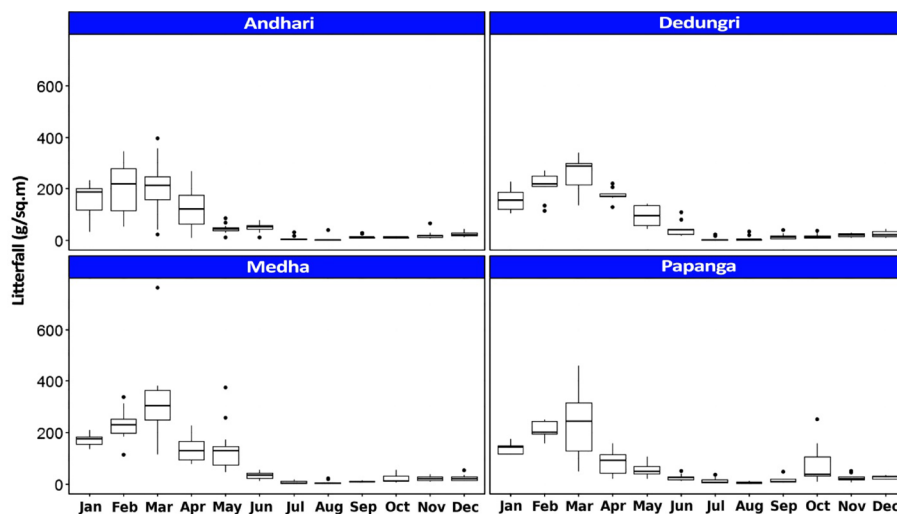


Fig. 3 Total annual litterfall production during the study period in the four sacred forests

Table 1 Physicochemical attributes (0–30 cm depth) of soils in four sacred forests

Soil parameters	Andhari	Dedungri	Medha	Papanga	p-value	Significance
pH	5.81	5.87	5.94	5.59	0.07622	NS
Electron conductivity	0.2	0.19	0.13	0.21	0.2498	NS
Moisture (%)	6.08	5.81	6.40	6.67	0.9679	NS
Carbon (%)	2.41	2.00	2.08	0.95	0.009408	***
Nitrogen (%)	0.17	0.16	0.15	0.06	0.002591	***
Phosphorus (Kg/ha)	7.61	14.12	14.52	38.48	3.36e-06	***
Potassium (Kg/ha)	197.75	195.61	192.66	199.85	0.9349	NS
C: N	14.87	13.50	14.47	14.77	0.6328	NS

Table 2 Annual litterfall and decomposition rates in the sacred forests

Parameters	Andhari	Dedungri	Medha	Papanga
Annual litter fall ($\text{Mg ha}^{-1} \text{ yr}^{-1}$)	8.59 ± 0.79	10.28 ± 0.92	11.32 ± 1.12	8.94 ± 0.78
Mean annual standing litter (X_L) (Mg ha^{-1})	2.39	2.51	2.68	2.47
Litter turnover rate (K_L)	3.59	4.10	4.22	3.62
Turnover time in year (T)	0.278	0.244	0.237	0.276
Residence time in days	102	89	86	101
Decay rate coefficient (k)	6.65	6.12	5.75	6.65
Time required for 50% decomposition in years (t_{50})	0.10	0.11	0.12	0.10
Time required for 99% decomposition in years (t_{99})	0.75	0.82	0.87	0.75

(T) ranged from 87 to 102 days across the four sacred forest sites and followed the order: Medha < Dedungri < Papanga < Andhari (Table 2).

3.3 Litter decomposition and mass loss

The in-situ litterbag experiment revealed a distinct phased pattern of decomposition. Litter mass loss increased progressively over time, although the rate of decomposition varied among sampling intervals. Mass loss was highest during the first month, declined in the second month, and increased again from the third month onward. Litter mass loss did not differ significantly among the four sacred forests ($p = 0.9079$). Approximately 95% of the initial litter mass was lost within six months across all study sites (Fig. 4). The decay rate coefficient (k) ranged from 5.75 to 6.65 yr^{-1} and followed the order: Andhari = Papanga > Dedungri > Medha. The estimated litter half-life (t_{50}) ranged from 0.10 to 0.12 years (38–44 days), while the time required for 99% decomposition (t_{99}) ranged from 0.75 to 0.87 years (274–318 days). Because the decomposition experiment was conducted for only six months, the t_{99} values represent model-based extrapolations derived from the fitted exponential decay model and should therefore be interpreted as estimated rather than directly observed decomposition times.

3.4 Initial biochemical properties of leaf litter

The initial chemical composition of leaf litter differed significantly among the four sacred forests for all measured nutrients except carbon (Table 3; Fig. 5). Papanga exhibited the highest concentrations of N, P, and K and the lowest C ratio, indicating comparatively higher nutrient concentrations and more favorable nutrient stoichiometric characteristics based on the parameters measured in this study. Based on these measured nutrient and stoichiometric characteristics, the sites were ranked as Papanga > Dedungri > Andhari > Medha. However, this interpretation should be considered with caution because structural and biochemical components such as lignin and cellulose, which are

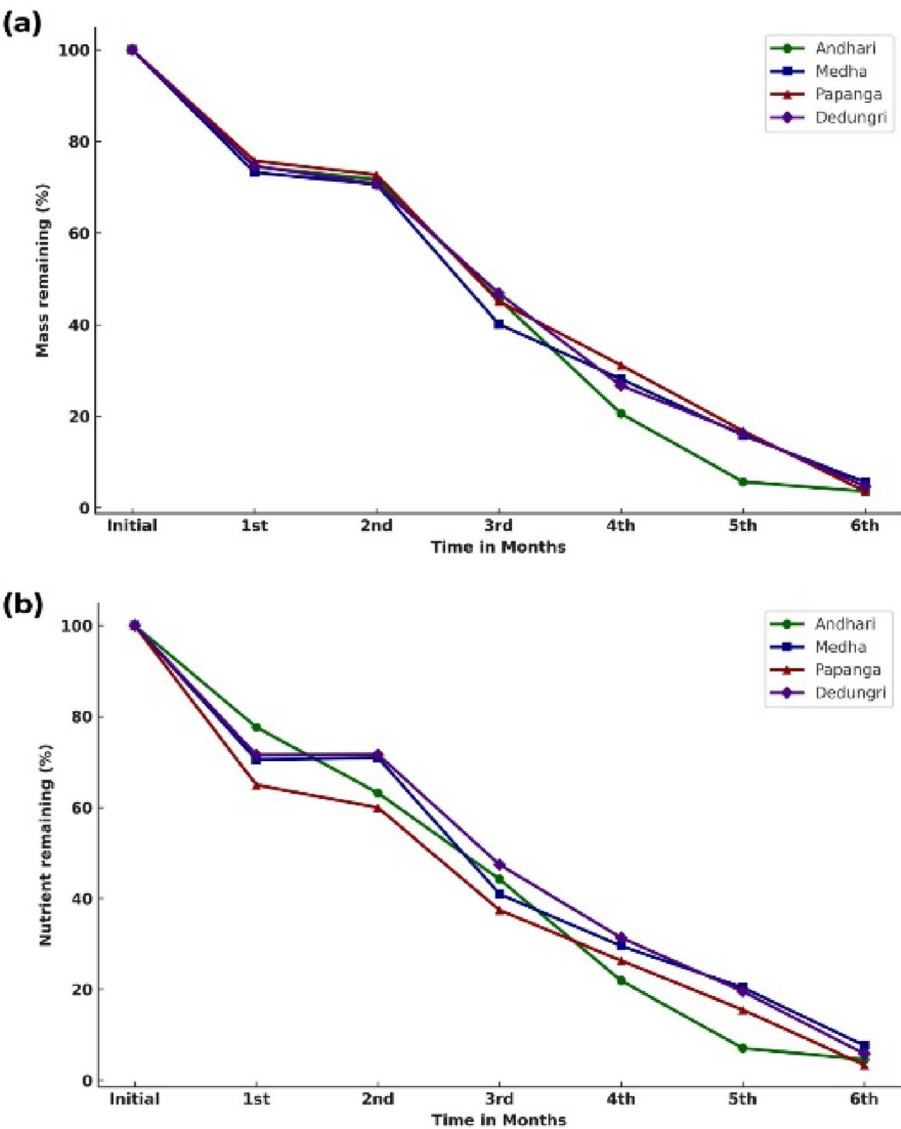


Fig. 4 Litter decomposition dynamics in the four sacred forests: **a** temporal changes in litter mass remaining, and **b** corresponding changes in nutrient content remaining in the litter

Table 3 Initial biochemical properties of leaf litter in four sacred forests

Leaf properties	Andhari	Dedungri	Medha	Papanga	p-value	Significance
%C	47 ± 0.63	46.67 ± 0.52	46.67 ± 0.52	46 ± 1.55	0.5159	NS
%N	0.59 ± 0.03	0.63 ± 0.07	0.54 ± 0.05	0.68 ± 0.06	0.006546	***
%P	0.26 ± 0.08	0.31 ± 0.02	0.18 ± 0.02	0.71 ± 0.03	0.0001977	***
%K	0.36 ± 0.05	0.44 ± 0.05	0.34 ± 0.07	0.75 ± 0.03	0.0006696	***
C/N	79.58 ± 3.69	74.78 ± 7.97	87.17 ± 9.48	68.22 ± 5.60	0.006239	***
C/P	192.19 ± 49.39	152.96 ± 8.83	268.72 ± 26.73	64.65 ± 3.86	0.0001977	***
N/P	2.41 ± 0.59	2.07 ± 0.29	3.13 ± 0.59	0.95 ± 0.11	0.0004914	***

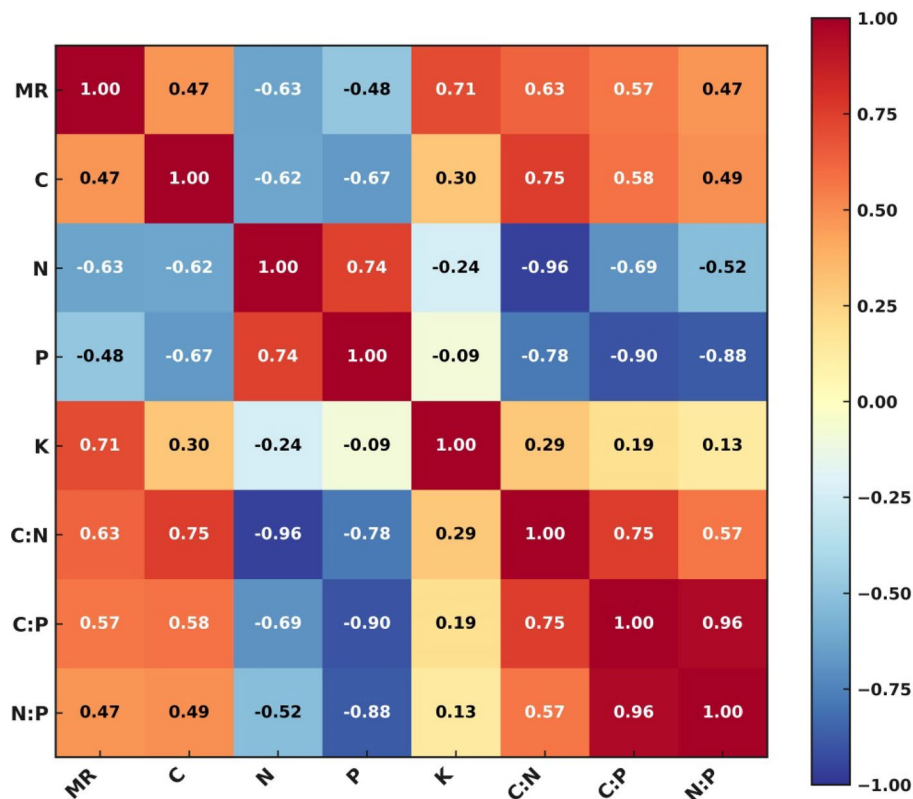


Fig. 5 Relationships between litter mass remaining and litter chemistry during decomposition period

important determinants of litter quality and decomposition, were not analyzed in the present study.

3.5 Nutrient return from litter decomposition

Nutrient release patterns closely followed the trends observed for litter mass loss (Fig. 6). Overall nutrient release did not differ significantly among the four sacred forests ($p = 0.996474$), whereas significant temporal variation was observed during the decomposition period ($p = 0.000136$). After six months of decomposition, Papanga recorded the highest total nutrient loss (96.57%), followed by Andhari (95.36%), Dedungri (94.18%), and Medha (92.38%). During decomposition, N and P concentrations increased in the residual litter, C remained relatively stable in Andhari and Medha but decreased in Dedungri and Papanga, whereas K declined markedly across all study sites (Fig. 7).

Nutrient-specific release patterns indicated that K exhibited the greatest release across all forests, followed by N and P. Potassium release was highest in Andhari (99.20%), followed by Papanga (99.09%), Medha (98.84%), and Dedungri (98.72%). Nitrogen loss followed the order Papanga (95.08%) > Andhari (95.06%) > Dedungri (93.58%) > Medha (92.26%), whereas phosphorus showed the lowest release, ranging from 95.34% in Papanga to 80.54% in Medha. No significant differences among forests were detected in the release of C ($p = 0.9836$), N ($p = 0.9778$), P ($p = 0.7221$), or K ($p = 0.9615$).

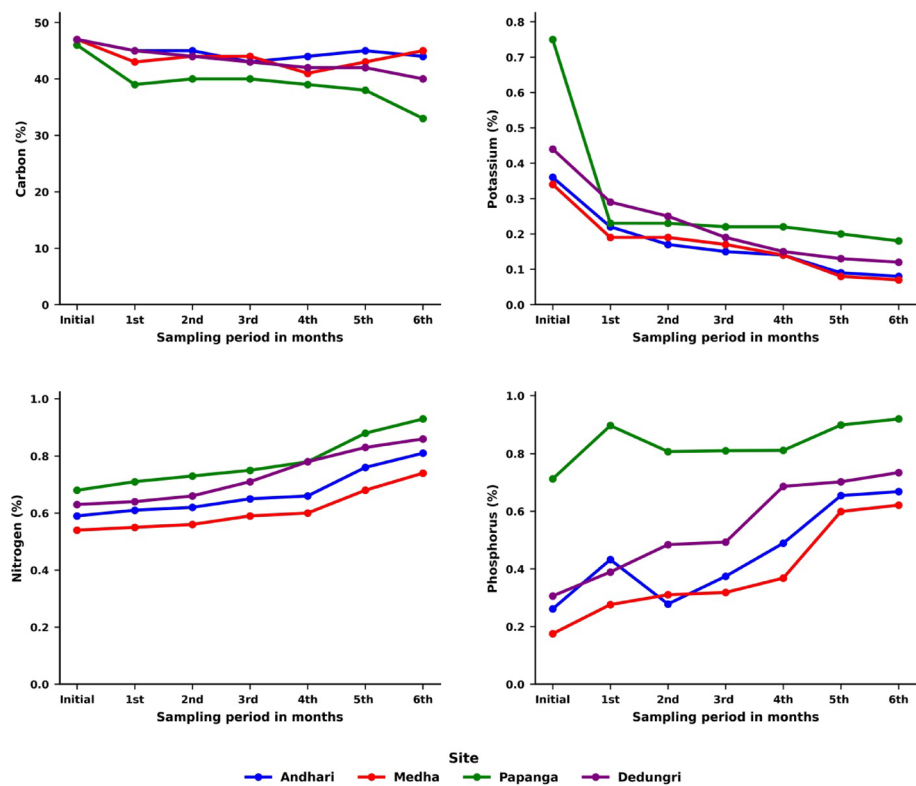


Fig. 6 Temporal patterns of nutrient loss and retention in litter during the decomposition period

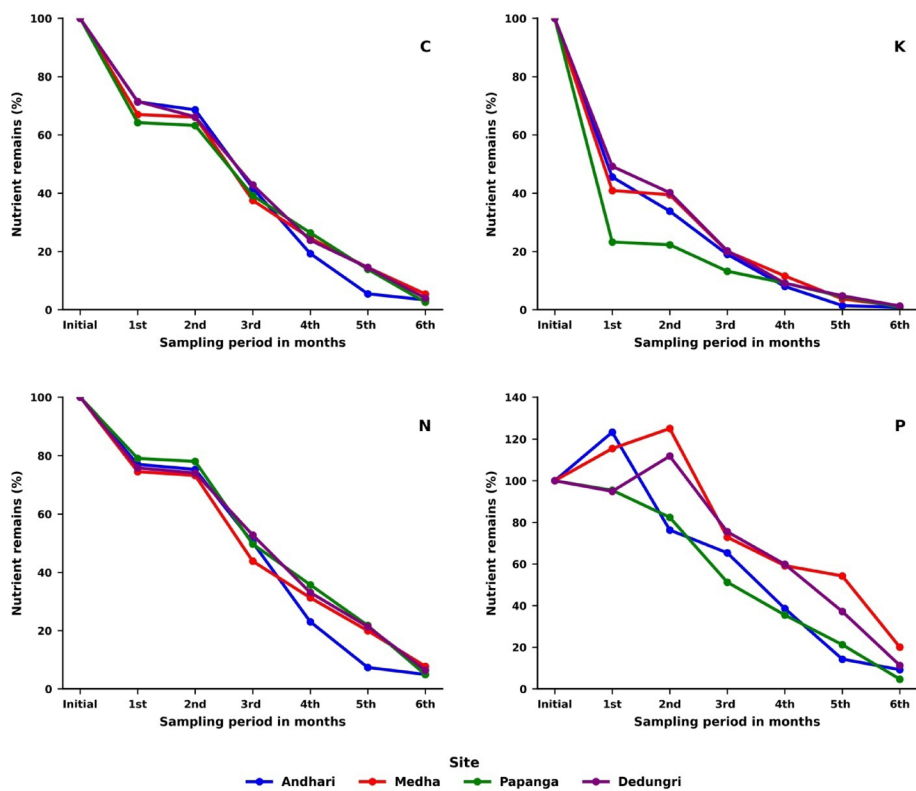


Fig. 7 Temporal dynamics of nutrient content in decomposing litter over six months

4 Discussion

Plant litter production is a major pathway linking vegetation productivity with soil organic matter formation and nutrient cycling [60–62]. In tropical dry deciduous sacred forests, the pronounced seasonal climate, particularly dry-season water stress, is likely to be a primary driver of litterfall dynamics by inducing leaf senescence and abscission as adaptive responses to seasonal moisture limitation [31]. The observed peak in litterfall during the dry season and the predominance of leaf litter (80–83% of total litterfall) are consistent with the deciduous phenology of tropical dry forests and indicate that leaf shedding is the principal pathway of litter input to the forest floor [23, 51, 63–67]. Although monthly litter production did not differ significantly among the four sacred forests, the pronounced seasonal variation suggests that temporal climatic and phenological factors exert a stronger influence on litterfall dynamics than site-level differences within these ecosystems. The dominance of leaf litter further indicates that seasonal nutrient transfer from vegetation to soil is primarily mediated through leaf turnover, with subsequent decomposition contributing substantially to nutrient release and soil fertility. Collectively, these findings emphasize the importance of seasonal litterfall dynamics in regulating nutrient cycling and ecosystem functioning in tropical dry deciduous sacred forests.

The magnitude of litterfall observed in these sacred forests falls within the range reported for tropical forest ecosystems globally [24]. However, annual litter production was higher than values previously reported for several tropical dry deciduous forests in India [29, 68]. This difference may be associated with variation in disturbance intensity, species composition, canopy structure, stand age, or site-specific environmental conditions, although these factors were not directly evaluated in the present study [27]. Future studies should explicitly examine the environmental and vegetation-related drivers of litter production across tropical dry deciduous sacred forests. In contrast, standing litter biomass was lower than that reported for several other tropical forest ecosystems in India [51, 69, 70]. Litter turnover rates were comparatively higher than those reported for tropical semi-deciduous [71, 72], tropical dry evergreen [51], and tropical evergreen forests [64], indicating relatively rapid litter turnover in these sacred forests. Such rapid turnover may facilitate faster nutrient recycling and organic matter processing within these ecosystems. The turnover rates, turnover times, and residence times reported in this study were derived from observed litterfall and standing litter measurements; however, confidence intervals or other uncertainty estimates were not calculated for these derived parameters. Therefore, these values should be interpreted as site-level estimates, and future studies incorporating uncertainty propagation and longer-term monitoring would provide a more robust assessment of variability in litter turnover and residence time.

Litter decomposition patterns further support the rapid turnover observed in these sacred forests. Although decomposition rates did not differ significantly among the four study sites ($p = 0.9923$), nearly 95% of the initial litter mass was lost within six months, a rate that is relatively high compared with values reported for several tropical forest ecosystems [73–78]. Similar patterns have been documented under tropical dry deciduous conditions, where more than 85% litter mass loss was observed within six months for home garden tree species [79–81]. The rapid decomposition observed in the present study is likely associated with favorable temperature and moisture conditions during

the decomposition period, which can enhance microbial activity and accelerate organic matter breakdown [82, 83]. In addition, variation in litter quality and forest disturbance may influence decomposition dynamics and nutrient cycling processes in tropical dry ecosystems [84]. Collectively, these results suggest that rapid litter decomposition may contribute substantially to nutrient recycling and organic matter turnover in tropical dry deciduous sacred forests, although the specific microbial and biochemical mechanisms were not directly examined in the present study.

Correlation analyses revealed strong associations between litter chemistry and decomposition dynamics. In addition to nutrient concentrations, structural components such as lignin and cellulose are widely recognized as important regulators of litter decomposition because they influence the recalcitrance of organic matter and its accessibility to decomposer microorganisms. High lignin concentrations generally slow decomposition due to their resistance to microbial degradation, whereas cellulose is more readily decomposed and contributes substantially to early-stage mass loss. Although lignin and cellulose were not quantified in the present study, variation in decomposition rates and nutrient release among sites may partially reflect differences in litter chemical quality and structural composition. Previous studies have shown that interactions between nutrient concentrations (particularly N and P) and structural compounds such as lignin, cellulose, and hemicellulose strongly influence decomposition dynamics in tropical ecosystems. Consequently, the rapid decomposition observed in these sacred forests may be associated with the combined effects of favorable climatic conditions, litter nutrient concentrations, and potentially lower lignin-to-nitrogen ratios in mixed-species litter, although this mechanism could not be directly evaluated in the present study.

Nitrogen and phosphorus concentrations were negatively correlated with litter mass remaining, whereas C, K, C, C, and N ratios showed positive correlations. The progressive increase in N and P concentrations in residual litter likely reflects microbial immobilization and nutrient accumulation during decomposition, a pattern widely reported in tropical forest ecosystems [7, 34, 85–88]. In contrast, K declined rapidly throughout decomposition, most likely due to its high solubility and susceptibility to leaching losses [87, 88]. The observed order of nutrient release ($K > N > P$) is consistent with patterns reported for other tropical forest ecosystems [72, 89, 90]. The rapid release of K reflects its high mobility, whereas N and P are more strongly regulated by microbial immobilization and mineralization processes. Overall, the results indicate that the sacred forests of Western Odisha are characterized by high litter production, rapid decomposition, and substantial nutrient turnover, highlighting their important role in nutrient cycling and ecosystem functioning in tropical dry deciduous landscapes.

The present study has some limitations that should be acknowledged. First, the absence of non-sacred forest control sites limits direct comparison of litterfall, decomposition, and nutrient fluxes between sacred and surrounding non-sacred forests. Because the study was conducted exclusively within sacred forests that differ in vegetation structure, species composition, and traditional management practices, a single non-sacred control site would not have provided an ecologically comparable baseline and could have introduced additional confounding factors. Consequently, the study focuses on seasonal and site-level variation within sacred forest systems rather than the specific effects of sacred-forest protection. Second, lignin and cellulose concentrations were not measured, although these structural components are important determinants of litter

decomposition rates and nutrient release patterns. Third, repeated temporal observations from the same forest sites were not explicitly analyzed using repeated-measures or mixed-effects models; therefore, temporal dependence among observations may not have been fully accounted for, and the correlation analyses should be interpreted as indicating associations rather than causal relationships. Future studies incorporating carefully matched sacred and non-sacred forest sites, lignin and cellulose analyses, mixed-effects or regression-based decomposition models, and longer-term monitoring would provide a more comprehensive understanding of the drivers of litter dynamics and nutrient cycling in sacred forests.

5 Conclusion

This study demonstrates that the sacred forests of Western Odisha are characterized by substantial litter production, rapid litter decomposition, and considerable nutrient turnover, highlighting their important role in nutrient cycling and ecosystem functioning in tropical dry deciduous landscapes. Annual litterfall ranged from 8.59 to 11.32 Mg ha⁻¹ yr⁻¹, with leaves contributing the dominant fraction of total litterfall. Decomposition was rapid, with decay rate constants ($k = 5.75\text{--}6.65$ yr⁻¹) and approximately 95% litter mass loss within six months, indicating relatively fast organic matter turnover. Nutrient release followed the order $K > N > P$, with initial litter nitrogen and phosphorus concentrations significantly associated with decomposition dynamics, while potassium loss was primarily influenced by leaching. These findings indicate that sacred forests contribute substantially to nutrient turnover and soil nutrient replenishment despite varying degrees of anthropogenic influence. The study was limited by the absence of non-sacred forest control sites, lignin and cellulose analyses, and statistical models that explicitly accounted for temporal dependence among repeated observations. Future studies integrating litter biochemical quality, microbial community dynamics, environmental drivers, and long-term monitoring would provide a more comprehensive understanding of decomposition and nutrient cycling in sacred forests. Overall, this study provides an important ecological baseline for the sacred forests of Western Odisha and underscores their significance as functionally important ecosystems for nutrient cycling, biodiversity conservation, and the long-term ecological sustainability of tropical dry deciduous landscapes.

Acknowledgements

The authors would like to express their gratitude to the Department of School of Life Sciences, Sambalpur University for providing essential lab facilities. Also, the authors would like to thank the local community and temple trusts for providing necessary and constant help and support during the time of field work.

Author contributions

Antaryami Pradhan contributed to the conceptualization, methodology design, data collection, data curation, and drafting of the original manuscript. Abinash Mansingh was involved in conceptual development, map preparation, statistical analysis, and critical revision and finalization of the manuscript. Jadhav Sudarshan Gopinath provided inputs on project design, scientific guidance, and value addition throughout the study. Nirius Jenan Ekka oversaw overall project supervision and provided logistical support. All authors have read and approved the final version of the manuscript.

Funding

No funds, grants, or other support were received during the preparation of this manuscript.

Data availability

The datasets generated and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and Consent to participate

Not Applicable.

Consent for publication

Not applicable.

Permissions to collect the plants/plant parts

As the study was based exclusively on collection of naturally fallen litter materials from the forest floor and did not involve removal of live plant material, no formal government permit was required. However, prior verbal consent was obtained from the respective local authorities, temple trust members, community heads, and/or head priests before conducting field activities within the sacred forest areas. The matter regarding permission for litter sample collection was discussed with the concerned Forest Department officials, who clarified that no specific approval is required for undertaking the proposed sampling.

Plant guidelines

The study was conducted following relevant local and national environmental guidelines and with full respect for local customs, traditional beliefs, and religious sentiments associated with the sacred forests. This is also a part of broader research on biocultural conservation and required consent and other information is traced from Pradhan and Ormsby, 2020 (Doi: <https://doi.org/10.1017/S0376892920000181>). The investigation involved only non-destructive litter sampling and caused no disturbance to the natural ecosystem.

Clinical trial number

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 21 April 2026 / Accepted: 17 September 2026

Published online: 20 September 2026

References

1. Vitousek PM. Litterfall, nutrient cycling, and nutrient limitation in tropical forests. *Ecology*. 1984;65:285–98. <https://doi.org/10.2307/1939481>.
2. Lowman MD. Litterfall and leaf decay in three Australian rainforest formations. *J Ecol*. 1988;76:451–65. <https://doi.org/10.2307/2260605>.
3. Pradhan A, Ormsby AA. Biocultural conservation in the sacred forests of Odisha, India. *Environ Conserv*. 2020;47(3):190–6. <https://doi.org/10.1017/S0376892920000181>.
4. Sayer EJ. Using experimental manipulation to assess the roles of leaf litter in the functioning of forest ecosystems. *Biol Rev Camb Philos Soc*. 2006;81:1–31. <https://doi.org/10.1017/S1464793105006846>.
5. Zhu X, Liu W, Chen H, Deng Y, Chen C, Zeng H. Effects of forest transition on litterfall, standing litter and related nutrient returns: implications for forest management in tropical China. *Geoderma*. 2019;333:123–34. <https://doi.org/10.1016/j.geoderma.2018.07.023>.
6. Williams-Linera G, Bonilla-Moheno M, López-Barrera F, Tolome J. Litterfall, vegetation structure and tree composition as indicators of functional recovery in passive and active tropical cloud forest restoration. *Ecol Manag*. 2021;493:119260. <https://doi.org/10.1016/j.foreco.2021.119260>.
7. Bargali SS, Singh SP, Singh RP. Pattern of weight loss and nutrients release from decomposing leaf litter in age series of Eucalyptus plantations. *Soil Biol Biochem*. 1993;25:1731–8. [https://doi.org/10.1016/0038-0717\(93\)90177-D](https://doi.org/10.1016/0038-0717(93)90177-D).
8. Rawat N, Nautiyal BP, Nautiyal MC. Litter decomposition rate and nutrient release from different litter forms in a Himalayan alpine ecosystem. *Environmentalist*. 2010;30(3):279–88. <https://doi.org/10.1007/s10669-010-9275-8>.
9. Sun J, Fu Y, Hu W, Bo Y, Nawaz M, Javed Q, Khattak WA, Akbar R, Xiaoyan W, Liu W, Du D. Early allelopathic input and later nutrient addition mediated by litter decomposition of invasive *Solidago canadensis* affect native plant and facilitate its invasion. *Front Plant Sci*. 2024;15:1503203. <https://doi.org/10.3389/fpls.2024.1503203>.
10. Jaramillo VJ, Murray-Tortorolo G, Martínez-Yrizar A, et al. Divergent litterfall nutrient responses to rainfall seasonality revealed through long-term observations in a tropical dry forest. *J Trop Ecol*. 2023;39:e25. <https://doi.org/10.1017/S0266467423000159>.
11. Watanabe T, Fukuzawa K, Shibata H. Temporal changes in litterfall, litter decomposition and their chemical composition in Sasa dwarf bamboo in a natural forest ecosystem of northern Japan. *J Res*. 2013;18(2):129–38. <https://doi.org/10.1007/s10310-011-0330-1>.
12. Campos CA, Cruz HL, Rocha OS. Mass, nutrient pool, and mineralization of litter and fine roots in a tropical mountain cloud forest. *Sci Total Environ*. 2017;575:876–86. <https://doi.org/10.1016/j.scitotenv.2016.09.126>.
13. Tian L, Zhao L, Wu X, Fang H, Zhao Y, Yue G, Liu G, Chen H. Vertical patterns and controls of soil nutrients in alpine grassland: implications for nutrient uptake. *Sci Total Environ*. 2017;607:855–64. <https://doi.org/10.1016/j.scitotenv.2017.07.080>.
14. Dai S, Wei T, Tang J, Xu Z, Gong H. Temporal changes in litterfall and nutrient cycling from 2005–2015 in an evergreen broad-leaved forest in the Ailao Mountains, China. *Plants*. 2023;12(6):1277. <https://doi.org/10.3390/plants12061277>.
15. Molofsky J, Augspurger CK. The effect of leaf litter on early seedling establishment in a tropical forest. *Ecology*. 1992;73(1):68–77. <https://doi.org/10.2307/1938721>.
16. Chaturvedi RK, Pandey SK, Tripathi A. Disturbance Mediated Changes in Litter Turnover and Nutrient Use Efficiency Facilitate Vegetation Shifts in Tropical Dry Ecosystems: Insights From a 10-Year Vegetation Management Study. *Land Degrad Dev*. 2026;37(8):3652–72. <https://doi.org/10.1002/ldr.70334>.

17. Facelli JM, Pickett STA. Plant litter: its dynamics and effects on plant community structure. *Bot Rev*. 1991;57(1):1–32. <https://doi.org/10.1007/BF02858763>.
18. Cuevas E, Lugo AE. Dynamics of organic matter and nutrient return from litterfall in stands of ten tropical tree plantation species. *Ecol Manag*. 1998;112:263–79. [https://doi.org/10.1016/S0378-1127\(98\)00410-1](https://doi.org/10.1016/S0378-1127(98)00410-1).
19. Zalamea M, González G. Leaf-fall phenology in a subtropical wet forest in Puerto Rico: from species to community patterns. *Biotropica*. 2008;40:295–304. <https://doi.org/10.1111/j.1744-7429.2007.00389.x>.
20. Becker J, Pabst H, Mnyonga J, Kuzyakov Y. Annual litterfall dynamics and nutrient deposition depending on elevation and land use at Mt. Kilimanjaro *Biogeosciences*. 2015;12:5635–46. <https://doi.org/10.5194/bg-12-5635-2015>.
21. Schilling EM, Waring BG, Schilling JS, Powers JS. Forest composition modifies litter dynamics and decomposition in regenerating tropical dry forest. *Oecologia*. 2016;182:287–97. <https://doi.org/10.1007/s00442-016-3662-x>.
22. Wagner FH, Hérault B, Bonal D, Stahl C, Anderson LO, Baker TR, et al. Climate seasonality limits leaf carbon assimilation and wood productivity in tropical forests. *Biogeosciences*. 2016;13:2537–62. <https://doi.org/10.5194/bg-13-2537-2016>.
23. Nakagawa M, Ushio M, Kume T, Nakashizuka T. Seasonal and long-term patterns in litterfall in a Bornean tropical rainforest. *Ecol Res*. 2019;34(1):31–9. <https://doi.org/10.1111/1440-1703.1003>.
24. Souza JR, Veloso MDM, Espírito-Santo MM, Silva JO, Sanchez-Azofeifa A, Brito BGS, Fernandes GW. Litterfall dynamics along a successional gradient in a Brazilian tropical dry forest. *Ecosyst*. 2019;6:35. <https://doi.org/10.1186/s40663-019-0194-y>.
25. Liu X, Feng Y, Zhao X, Cui Z, Liu P, Chen X, Zhang Q, Liu J. Climatic drivers of litterfall production and its components in two subtropical forests in South China: a 14-year observation. *Agric Meteorol*. 2024;344:109798. <https://doi.org/10.1016/j.agrfo rmet.2023.109798>.
26. Barlow J, Gardner TA, Ferreira LV, Peres CA. Litter fall and decomposition in primary, secondary and plantation forests in the Brazilian Amazon. *Ecol Manag*. 2007;247:91–7. <https://doi.org/10.1016/j.foreco.2007.04.017>.
27. Chaturvedi RK, Tomlinson KW, Pandey SK, Tripathi A, Raghubanshi AS, Singh JS. Changes in plant traits and productivity of two functional types across a soil water content gradient in a tropical dry forest. *J Ecol*. 2025;113:3535–49. <https://doi.org/10.1111/1365-2745.70177>.
28. Zhang H, Yuan W, Dong W, Liu S. Seasonal patterns of litterfall in forest ecosystems worldwide. *Ecol Complex*. 2014;20:240–7. <https://doi.org/10.1016/j.ecocom.2014.01.003>.
29. Yang Y, Yang H, Wang Q, Dong Q, Yang J, Wu L, You C, Hu J, Wu Q. Effects of two management practices on monthly litterfall in a cypress plantation. *Forests*. 2022;13(10):1581. <https://doi.org/10.3390/f13101581>.
30. Lopes MCA, Araújo VFP, Vasconcellos A. The effects of rainfall and vegetation on litterfall production in the semiarid region of northeastern Brazil. *Braz J Biol*. 2015;75:703–8. <https://doi.org/10.1590/1519-6984.21613>.
31. Magh T, Mozhui L, Kakati LN, Ao B, Lemtur T, Jing L. Litter decomposition and nutrient dynamics in a subtropical ecosystem: a comparison of natural and plantation forests (*Duabanga grandiflora*) in Nagaland, North-East India. *Glob Ecol Conserv*. 2024;56:e03321. <https://doi.org/10.1016/j.gecco.2024.e03321>.
32. Vendrami JL, Jurinitz CF, Castanho CT, Lorenzo L, Oliveira AA. Litterfall and leaf decomposition in forest fragments under different successional phases on the Atlantic Plateau of the state of São Paulo, Brazil. *Biota Neotrop*. 2012;12. Available from: <https://www.redalyc.org/pdf/1991/199124391016.pdf>
33. Rai A, Singh AK, Ghosal N, Singh N. Understanding the effectiveness of litter from tropical dry forests for the restoration of degraded lands. *Ecol Eng*. 2016;93:76–81. <https://doi.org/10.1016/j.ecoleng.2016.05.014>.
34. Giweta M. Role of litter production and its decomposition, and factors affecting the processes in a tropical forest ecosystem: a review. *J Ecol Environ*. 2020;44(1):11. <https://doi.org/10.1186/s41610-020-0151-2>.
35. Gessner MO, Swan CM, Dang CK, McKie BG, Bardgett RD, Wall DH, Hättenschwiler S. Diversity meets decomposition. *Trends Ecol Evol*. 2010;25:372–80.
36. Rottmann N, Dyckmans J, Joergensen RG. Microbial use and decomposition of maize leaf straw incubated in packed soil columns at different depths. *Eur J Soil Biol*. 2010;46(1):27–33. <https://doi.org/10.1016/j.ejsobi.2009.09.004>.
37. Li T, Ye Y. Dynamics of decomposition and nutrient release of leaf litter in *Kandelia obovata* mangrove forests with different ages in Jiulongjiang Estuary, China. *Ecol Eng*. 2014;73:454–60. <https://doi.org/10.1016/j.ecoleng.2014.09.102>.
38. MacLean DA, Wein RW. Weight loss and nutrient changes in decomposing litter and forest floor material in New Brunswick forest stands. *Can J Bot*. 1978;56(21):2730–49. <https://doi.org/10.1139/b78-326>.
39. Moore TR, Trofymow JA, Prescott CE, Fyles J, Titus BD. Patterns of carbon, nitrogen and phosphorus dynamics in decomposing foliar litter in Canadian forests. *Ecosystems*. 2006;9:46–62. <https://doi.org/10.1007/s10021-004-0026-x>.
40. Krishna MP, Mohan M. Litter decomposition in forest ecosystems: a review. *Energy Ecol Environ*. 2017;2(4):236–49.
41. Rawat M, Jägerbrand AK, Bai Y, Alatalo JM. Litter decomposition above the treeline in alpine regions: a mini review. *Acta Oecol*. 2021;113:103775. <https://doi.org/10.1016/j.actao.2021.103775>.
42. Trofymow JA, Moore TR, Titus BD, et al. Rates of litter decomposition over 6 years in Canadian forests: influence of litter quality and climate. *Can J Res*. 2002;32:789–804. <https://doi.org/10.1139/x01-117>.
43. Fierer N, Craine JM, McLaughlan K, Schimel JP. Litter quality and the temperature sensitivity of decomposition. *Ecology*. 2005;86(2):320–6. <https://doi.org/10.1890/04-1254>.
44. Davidson EA, Janssens IA. Temperature sensitivity of soil carbon decomposition and feedbacks to climate change. *Nature*. 2006;440(7081):165–73. <https://doi.org/10.1038/nature04514>.
45. Cornwell WK, Cornelissen JHC, Amatangelo K, et al. Plant species traits are the predominant control on litter decomposition rates within biomes worldwide. *Ecol Lett*. 2008;11:1065–71. <https://doi.org/10.1111/j.1461-0248.2008.01219.x>.
46. Prescott C. Litter decomposition: what controls it and how can we alter it to sequester more carbon in forest soils? *Biogeochemistry*. 2010;101:133–49. <https://doi.org/10.1007/s10533-010-9439-0>.
47. Berg B, McLaugherty C. Plant litter: decomposition, humus formation, carbon sequestration. Berlin: Springer; 2014. https://doi.org/10.1007/978-3-540-74923-3_6.
48. Singh NS, Brearley FQ, Tripathi SK. Climate control of litter decomposition and nutrient release in tropical and sub-tropical forest biomes of Northeast India. *Environ Adv*. 2025;100634. <https://doi.org/10.1016/j.envadv.2025.100634>.
49. Deswal S. Sacred groves as traditional ecological heritage and conservation model for sustainable development. 2024. ISSN: 10123407.
50. Khiwtam RS, Ramakrishnan PS. Litter and fine root dynamics of a relict sacred grove forest at Cherrapunji in north-eastern India. *Ecol Manag*. 1993;60:327–44. [https://doi.org/10.1016/0378-1127\(93\)90087-4](https://doi.org/10.1016/0378-1127(93)90087-4).

51. Pragasam LA, Parthasarathy N. Litter production in tropical dry evergreen forests of south India in relation to season, plant life forms and physiognomic groups. *Curr Sci*. 2005;88(8):1255–63.
52. Pradhan A, Ormsby AA, Behera N. A comparative assessment of tree diversity, biomass and biomass carbon stock between a protected area and a sacred forest of Western Odisha, India. *Ecoscience*. 2019;26(3):195–204. <https://doi.org/10.1080/11956860.2019.1586118>.
53. Crossley DA, Hoglund MP. A litter bag method for the study of microarthropods inhabiting leaf litter. *Ecology*. 1962;43:572–3.
54. Reiners WA, Reiners NM. Energy and nutrient dynamics of forest floors in three Minnesota forests. *J Ecol*. 1970;58:497–519.
55. Olson JS. Energy storage and the balance of producers and decomposers in ecological systems. *Ecology*. 1963;44(2):322–31. <https://doi.org/10.2307/1932179>.
56. Bockheim JG, Jepsen EA, Heisey DM. Nutrient dynamics in decomposing leaf litter of four tree species on a sandy soil in northwestern Wisconsin. *Can J Res*. 1991;21:803–12. <https://doi.org/10.1139/x91-113>.
57. Allen SE. Chemical analysis of ecological materials. 2nd ed. Oxford: Blackwell Scientific; 1989.
58. Allen SE, Grimshaw HM, Parkinson JA, Quamby C. Chemical analysis of ecological materials. Oxford: Blackwell Scientific; 1974.
59. Jackson ML. Soil chemical analysis. Englewood Cliffs, NJ: Prentice-Hall; 1958. pp. 146–51.
60. Swift MJ, Heal OW, Anderson JM. Decomposition in terrestrial ecosystems. Oxford: Blackwell; 1979.
61. Bossa JR, Adams JF, Shannon DA, Mullins GL. Phosphorus and potassium release pattern from *Leucaena* leaves in three environments of Haiti. *Nutr Cycl Agroecosyst*. 2005;73:25–35. <https://doi.org/10.1007/s10705-005-7127-6>.
62. Liu S, Yang R, Peng X, Hou C, Ma J, Guo J. Contributions of plant litter decomposition to soil nutrients in ecological tea gardens. *Agriculture*. 2022;12(7):957. <https://doi.org/10.3390/agriculture12070957>.
63. Robertson GP, Paul EA. Decomposition and soil organic matter dynamics. *Methods in ecosystem science*. New York, NY: Springer; 2000. pp. 104–16. https://doi.org/10.1007/978-1-4612-1224-9_8.
64. Rajendraprasad M, Krishnan PN, Pushpangadan P. Vegetational characterisation and litter dynamics of the sacred groves of Kerala, southwest India. *J Trop Sci*. 2000;12(2):320–35.
65. Barbosa JHC, Faria SM. Aporte de serrapilheira ao solo em estágios sucessionais florestais na Reserva Biológica de Poço das Antas, Rio de Janeiro Brasil. *Rodriguésia*. 2006;57(3):461–76. <https://doi.org/10.1590/2175-7860200657306>.
66. Dickow KMC, et al. Produção de serrapilheira em diferentes fases sucessionais de uma floresta subtropical secundária, em Antonina. *PR Cerne*. 2012;18(1):75–86. <https://doi.org/10.1590/S0104-77602012000100010>.
67. Aguilar-Cruz Y, García-Franco JG, Zotz G. Litter-trapping tank bromeliads in five different forests: carbon and nutrient pools and fluxes. *Biotropica*. 2022;54(1):170–80. <https://doi.org/10.1111/btp.13048>.
68. Shukla G, Pala NA, Chakravarty S. Quantification of organic carbon and primary nutrients in litter and soil in a foothill forest plantation of eastern Himalaya. *J Res*. 2017;28(6):1195–202. <https://doi.org/10.1007/s11676-017-0394-7>.
69. Visalakshi N. Litterfall, standing crop of litter and their nutrients in two tropical dry evergreen forests in India. *Int J Ecol Environ Sci*. 1993;19:163–80.
70. Sundarapandian SM, Swamy PS. Litter production and leaf-litter decomposition of selected tree species in tropical forests at Kodayar in the Western Ghats, India. *Ecol Manag*. 1999;123:231–44. [https://doi.org/10.1016/S0378-1127\(99\)00062-6](https://doi.org/10.1016/S0378-1127(99)00062-6).
71. Sanches L, Valentini CMA, Junior OBP, et al. Seasonal and interannual litter dynamics of a tropical semideciduous forest of the southern Amazon Basin, Brazil. *J Geophys Res*. 2008;113:G04007. <https://doi.org/10.1029/2007JG000593>.
72. Arunachalam A, Maithani K, Pandey HN, Tripathi RS. Leaf litter decomposition and nutrient mineralization patterns in regrowing stands of a humid subtropical forest after tree cutting. *Ecol Manag*. 1998;109:151–61. [https://doi.org/10.1016/S0378-1127\(98\)00240-0](https://doi.org/10.1016/S0378-1127(98)00240-0).
73. Semwal RL, Maikhuri RK, Rao KS, Sen KK, Saxena KG. Leaf litter decomposition and nutrient release patterns of six multipurpose tree species of central Himalaya, India. *Biomass Bioenergy*. 2003;24:3–11. [https://doi.org/10.1016/S0961-9534\(02\)00087-9](https://doi.org/10.1016/S0961-9534(02)00087-9).
74. Isaac SR, Nair MA. Litter dynamics of six multipurpose trees in a homegarden in Southern Kerala, India. *Agrofor Syst*. 2006;67:203–13. <https://doi.org/10.1007/s10457-005-1107-3>.
75. Barbhuiya AR, Arunachalam A, Nath PC, Khan ML, Arunachalam K. Leaf litter decomposition of dominant tree species of Nandapha National Park, Arunachal Pradesh, northeast India. *J Res*. 2008;13(1):25–34. <https://doi.org/10.1007/s10310-007-0044-6>.
76. Wang Q, Wang S, Huang Y. Comparison of litter fall, litter decomposition and nutrient return in a monoculture *Cunninghamia lanceolata* and a mixed stand in southern China. *Ecol Manag*. 2008;255:1210–8. <https://doi.org/10.1016/j.foreco.2007.10.026>.
77. Das T, Das AK. Litter production and decomposition in the forested areas of traditional home gardens: a case study from Barak Valley, Assam, Northeast India. *Agrofor Syst*. 2010;79:157–70. <https://doi.org/10.1007/s10457-010-9284-0>.
78. Mehta N, Dinakaran J, Patel S, Laskar AH, Yadava MG, Ramesh R, Krishnayya NSR. Changes in litter decomposition and soil organic carbon in a reforested tropical deciduous cover (India). *Ecol Res*. 2013;28:239–48. <https://doi.org/10.1007/s11284-012-1011-z>.
79. Allison SD, Vitousek PM. Rapid nutrient cycling in leaf litter from invasive plants in Hawaii. *Oecologia*. 2004;141(4):612–9. <https://doi.org/10.1007/s00442-004-1679-z>.
80. Rawlik K, Nowiński M, Jagodziński AM. Short life–fast death: decomposition rates of woody plants leaf- and herb-litter. *Ann Sci*. 2021;78(1):6. <https://doi.org/10.1007/s13595-020-01019-y>.
81. Upadhyaya K, Sahoo UK, Vanlalhriatpuia K, Roy S. Decomposition dynamics and nutrient release pattern from leaf litters of five commonly occurring home garden tree species in Mizoram, India. *J Sustain For*. 2012;31(8):711–26. <https://doi.org/10.1080/10549811.2012.706495>.
82. Lavelle P, Blanchart A, Martin A, Spain F, Toutain F, Barois I, Schaeffer R. Hierarchical model for decomposition in terrestrial ecosystems: application to soils of the humid tropics. *Biotropica*. 1993;25(2):130–50.
83. Zhang L, Guo W, Chen Y, Li Z, Liu Q, Heal KV, Li S, Zhou L. Simulated warming increases litter decomposition and release rates of some metallic elements and recalcitrant components in different-aged Chinese fir plantations. *Forests*. 2024;15(12):2151. <https://doi.org/10.3390/f15122151>.

84. Leal F, Aburto F, Aguilera N, Echeverría C, Gatica-Saavedra P. Forest degradation modifies litter production, quality, and decomposition dynamics in southern temperate forests. *Front Soil Sci.* 2023;3:1111694. <https://doi.org/10.3389/fsoil.2023.1111694>.
85. Gosz JR, Likens GE, Bormann FH. Nutrient release from decomposing leaf and branch litter in the Hubbard Brook Forest, New Hampshire. *Ecol Monogr.* 1973;43(2):173–91. <https://doi.org/10.2307/1942193>.
86. Pandey U, Singh JS. Leaf litter decomposition in an oak–conifer forest in Himalaya: the effects of climate and chemical composition. *Forestry.* 1982;55:47–59. <https://doi.org/10.1093/forestry/55.1.47>.
87. Peterson DL, Rolfe GL. Nutrient dynamics of herbaceous vegetation in upland and flood plain forest communities. *Am Midl Nat.* 1982;107:325–39.
88. Liang G, Sun P, Waring BG, Fu Z, Reich PB. Alleviating nitrogen and phosphorus limitation does not amplify potassium-induced increase in terrestrial biomass. *Glob Change Biol.* 2025;31(4):e70193. <https://doi.org/10.1111/gcb.70193>.
89. Sarkar M, Devi A, Nath M. Foliar litter decomposition of four dominant tree species in the Hollongapar Gibbon Wildlife Sanctuary, Assam, North East India. *Curr Sci.* 2016;111(4):747–53.
90. Awasthi P, Bargali K, Bargali SS, Khatri K. Nutrient return through decomposing *Coriaria nepalensis* litter in degraded hills of Kumaun Himalaya, India. *Front Glob Change.* 2022;5:1008939. <https://doi.org/10.3389/ffgc.2022.1008939>.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.